

Harpreet Singh

Computer Engineering Student
Backend / Full-Stack Software

hs3732065@gmail.com | +1 (778) 998-1236

Vancouver, BC, Canada

linkedin.com/in/harpreet-singh-4523661a0 | github.com/harpreeto6

EDUCATION

University of British Columbia (UBC)
BASc, Computer Engineering (Vancouver, BC)

Sept 2024 to Dec 2027 (expected)

EXPERIENCE

Software Development Engineer Intern, Amazon

May 2026 to Jul 2026

- Developing software in a production engineering environment, contributing to implementation, testing, debugging, and code review workflows.
- Building maintainable backend/service functionality while working within a large-scale codebase and established engineering practices.

MOps Software Developer, UBC ORBIT Satellite Design Team

Jan 2026 to Present

- Building mission-operations software across frontend, backend, deployment, and testing workstreams.
- Working with other ORBIT subteams to turn engineering requirements into usable tools, API integrations, and reliable internal workflows.

PROJECTS

Sensum: Wellbeing + Productivity Platform

Personal Project

Java 17 | Spring Boot | PostgreSQL | Next.js | TypeScript | Tailwind | Chrome MV3

- Built a 3-part monorepo with a Spring Boot API, PostgreSQL database, Next.js web app, and MV3 browser extension.
- Implemented BCrypt password hashing and HttpOnly JWT auth; built REST routes across quests, events, stats, settings, achievements, friends, and buddy sessions.
- Designed product loops for 3 quest recommendations, complete/skip/snooze actions, XP/leveling, daily streaks, social features, and extension telemetry batching.

RAG Retrieval Pipeline

Academic Team Project

Python | PyMuPDF | all-MiniLM-L6-v2 | FAISS | JSONL | Unit Tests

- Built the retrieval layer for a computer-systems QA pipeline: PDF extraction, page cleaning, chunking, embedding, indexing, and top-k search.
- Chunked 8 textbook/source files into 11,890 MiniLM-ready chunks and generated 384-d embeddings for a normalized FAISS inner-product index.
- Implemented query-time retrieval that embeds questions, searches FAISS, and returns ranked source/page metadata for downstream answer generation.

AI Agents and MCP Exploration

Self-Directed Project

Google ADK | MCP | A2A | Tool Calling | Agent Workflows

- Built hands-on agent prototypes while studying Google Agent Development Kit concepts, MCP tool integration, and Agent-to-Agent communication patterns.
- Followed Hugging Face MCP course material to understand tool servers, external data/tool access, structured calls, and agent workflow design.

QuestBoard: Cloud-Deployed Quest Service

Personal Project

Java | Spring Boot | TypeScript | MCP | Docker | AWS CDK | ECS/Fargate

- Built a monorepo with a Spring Boot orchestrator and TypeScript MCP server connected through REST/MCP endpoints.
- Wrote Dockerized services and AWS CDK infrastructure for a VPC, ECS/Fargate cluster, load-balanced orchestrator, service discovery, and CloudWatch logs.

Canvas Assistant: AI Chatbot

Hackathon

Node.js | Google Dialogflow | Canvas API

- Built a student-support chatbot prototype using Dialogflow and Canvas-style data flows for real-time assignment/course responses.
- Focused on rapid API integration, intent handling, and structured conversational responses under hackathon time constraints.

SKILLS

Languages: Java, TypeScript, JavaScript, Python, SQL, C++

AI/Cloud: MCP, Google ADK, FAISS, AWS CDK, Docker

Backend: Spring Boot, REST APIs, Node.js, JWT, PostgreSQL

Frontend/Tools: React / Next.js, Tailwind, Chrome MV3, Git